

IY4113 Milestone 1

Assessment Details	Please Complete All Details
Group	4113 A
Module Title	Applied Software Engineering using object Orientated Programming
Assessment Type	Java fundamental's part 1
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Student ID Number	P512757
Date of Submission	03/05/26
Word Count	

I confirm that this assignment is my own work. Where I have referred to academic sources,

☒ *I have provided in-text citations and included the sources in the final reference list.*

☒ *Where I have used AI, I have cited and referenced appropriately.

Purpose of the Program

*Add purpose of the program, identify core program functionality and and system constraints.

The purpose of this program is to develop a console-based Java application called City Ride Lite. The goal is to assist a user in observing and tracking their public transport journeys for a single day. The program must store detailed information by asking the user to input data. Based on the dails , the fate and discounts will be calculated according to the zone crossed. It is also necessary to apply a daily cap and adjust the cost accordingly . The program is required to store all the journeys to allow the user to review and filter them after confirmation, as well as remove them if needed. A summary will be presented, including the total number of journeys, total amount spend, average cost per journey, and the most expensive journey.In addition, the program will provide clear menu options to guide the user and ensure correct input.

Core program functionality

Display the menu options***

- Show the options details(add journey, list journeys, filter journeys, view daily summary, view totals by passenger type, undo/remove journeys , reset day and exit)

Add journeys details***

- Capture the user input regarding start zone, destination zone, passenger types, and time band
- Assign a unique journey ID

System Validations

- Validate user input
- Prompt user to re-enter invalid input

Make fare calculations of each journey***

- Calculate the number of zones crossed and determine the base fare based on dataset
- Apply passenger discount : Adult (0%), Student(25%), Child(50%), Senior (30%)

Apply daily cap***

- Track running total per passenger type
- Ensure total does not exceed daily cap
- Adjust fare if cap is reached
- Maintain separate running totals for each passenger type
- Round all calculated fares to 2 decimal places

Display the summary of journeys details***

- Show the totals of daily summary
- Display the totals by passenger types
- Present the category counts
- Adjust the average cost
- Generate the most expensive journey

Manage journeys details***

- Display all recorded journeys
- Allow user to filter journeys based on criteria
- Allow user to remove a journey by ID

Reset system***

- Clear all stored journeys and totals
- Ask for confirmation before resetting

System constraints

- The program must be console-based only, using menu options for user interaction
- The system must store all journey data in memory during execution only, meaning data will be lost when the program is reset or closed
- Each journey must have a unique ID to support viewing and removing journeys
- The system must run continuously and display menu options until the user selects exit
- The system must handle invalid input and display error messages without crashing
- The system must correctly update data when journeys are removed or reset
- The system must maintain separate running totals for each passenger type
- The system must ensure all monetary values are rounded to 2 decimal places
- The system must correctly process fare calculations, including zones, discounts, and daily caps.

Input Process Output Table

Add IPO table (It maybe easier to create the table in Word and paste as an image!)

Input Process Output Table

Feature / Task	Inputs	Processing (What the system does)
Start program	None (program launch)	Initialise system; display main menu
Display main menu	User menu choice	Validate input; route to selected option
Add journey	Start zone, destination zone, passenger type, time band	Validate inputs; calculate zones crossed; determine base fare; apply discount; assign unique ID; store journey
Validate input	User input values	Check zones (1-5), valid passenger type, valid time band; show error if invalid
Calculate fare	Valid journey details	Calculate zones crossed; get base fare from dataset; apply discount
Apply daily cap	Running total per passenger	Check if total exceeds cap; adjust fare if needed; update running total
Store journey	Journey details	Save journey in memory with unique ID
List journeys	None/ user selection	Retrieve and display all stored journeys
Filter journeys	Filter criteria (type, time, zone, date)	Search stored journeys; display matching results
Remove journey	Journey ID	Find journey; remove it; update totals
Reset system	User confirmation	Clear all stored journeys and totals
Display summary	Stored journey data	Calculate total journeys, total cost, average cost, most expensive journey
Display totals by passenger type	Stored journey data	Calculate totals per passenger type
Display category counts	Stored journey data	Count peak vs off-peak; zone usage
Handle invalid input	Incorrect user input	Display error message; prompt again
Exit program	Menu choice	Terminate program

Gantt Chart

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Java project

Project start: Mon, 4/27/2026

Display week: 1

SIMPLE GANTT CHART by Vertex42.com

https://www.vertex42.com/ExcelTemplates/simple-gantt-chart.html

TASK	PROGRESS	START	END
Milestone 1 - Project Initiation & Planning			
Purpose of the program	100%	4/27/26	4/27/26
Core Functionality	100%	4/27/26	4/29/26
System Constraints	100%	4/29/26	4/29/26
IPO table	100%	5/1/26	5/1/26
Gantt Chart	100%	4/29/26	5/2/26
Diary Entries	100%	4/27/26	5/2/26
Milestone 2 - Research & System Design			
Research similar system	0%	5/4/26	5/10/26
Design program structure	0%	5/4/26	5/10/26
Create Flowchart	0%	5/4/26	5/10/26
Plan Classes (OOD design)	0%	5/4/26	5/10/26
Write Diary Entries	0%	5/4/26	5/10/26
Milestone 3 - Coding Implementation			
Setup project (IntelliJ)	0%	5/11/26	5/17/26
Implement journey input	0%	5/11/26	5/17/26
zones calculation	0%	5/11/26	5/17/26
Fare calculation (dataset)	0%	5/11/26	5/17/26
Apply discounts	0%	5/11/26	5/17/26
GitHub commits	0%	5/11/26	5/17/26
Write Diary Entries	0%	5/11/26	5/17/26
Milestone 4 - Testing, Refinement & Final Submission			
Filtering journeys	0%	5/18/26	5/24/26
Summaries (totals, averages)	0%	5/18/26	5/24/26
Remove/reset journeys	0%	5/18/26	5/24/26
Daily caps implementation	0%	5/18/26	5/24/26
Testing & debugging	0%	5/18/26	5/24/26
code cleanup (standards)	0%	5/18/26	5/24/26
GitHub updates	0%	5/18/26	5/24/26
Write diary	0%	5/18/26	5/24/26

Add diary entries here detailing what you have done, why you have done it, and any problems encountered.

****27/01/2026 - Daily Entry 1- Understanding the assignment brief , writing program purpose and functionality**

Today I reviewed the assignment brief to understand the requirements of the program. I identified the main features of the program neatly. This helped me understand what the system is expected to do. I did read the brief repeatedly to get the main idea. In addition, I wrote the purpose of the program breaking it down into each section, such as highlighting, collecting facts and analysing. I want to make sure which information I should put for each stage. After that, I defined the core functionality of the program. I wrote it down based on the brief and included the short under each subheading.

The problem I encountered was analysing the long brief as well as the details. It took me a few times reading that and I found it challenging to rewrite the purpose and the functionality in my own words without copying directly from the scenario. For the functionality, I initially found it difficult to understand how the daily caps should be applied. Well, I have not completed functionality fully and constraints today. I also missed some features in functionality, and I need some requirements to fix it correctly.

****29/01/2026 - Daily Entry 2 - Working on Gantt Chart, functionality and constraints**

Today, I continued working by focusing on the Gantt chart and reviewing my functionality again. I did some research for the Gantt chart template and what information needs to be included based on the milestone requirements. I also started creating my own Gantt chart. In addition, I made some small improvements to ensure the functionality section is finished. I also started thinking about the system constraints and how they are different from functionality. I realised that functionality explains what the system does, while constraints describe the rules and limitations of the system. This helped me understand the difference clearly.

The problem I encountered today was finding a suitable Gantt chart template, as I found it quite difficult to choose the correct format. I also found it difficult to choose the correct format. I also found it a bit confusing to clearly separate functionality and constraints at first, but after reviewing my work, I started to understand it better.

Overall, today's work was lighter compared to the first day, but still helped me make progress and improve my understanding of the assignment.

01/05/26 - Daily Entry 3 - IPO table and Gantt Chart

Today I continued working by focusing on the IPO table and the Gantt chart. I created the IPO table by identifying the inputs, processes, and outputs of the system based on the functionality I had written. This helped me understand how the system works step by step. I also started working on the Gantt chart using Excel.

The problem I encountered was structuring the IPO correctly and deciding how to organise the tasks in the Gantt Chart. At first, I was not sure how detailed it should be. However, after reviewing some examples, I was able to improve my understanding and make it clear.

02/05/26 - Daily Entry 4 - Finalising Milestone 1

Today I focused on reviewing and finalising . I checked my purpose of the program , functionality , constraints , IPO table , and Gantt chart to make sure everything is consistent and matches the requirements.

I made some small improvements to my work and ensured that all sections are clearly written and connected properly. I also reviewed my diary entries to make sure they reflect the progress I have made during this timeline.

The main problem I faced was making sure all sections were aligned and not repeated , especially between functionality and constraints .For example, I checked the fare calculations were clearly explained in functionality and not repeated in the constraints section.This is my first time completing this type of assignment , and although there were many requirements, I have done my best carefully complete and review all parts of the milestone.